

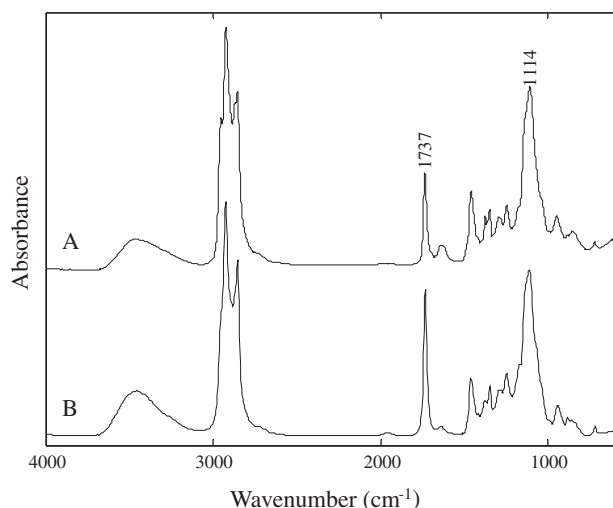


Fig. 5. Infrared spectra obtained from (A) the residue washed from the PP fibre surface with warm petroleum spirits and (B) a typical PEO saturated fatty acid ester.

is in very good agreement with that of a typical PEO saturated fatty acid ester which is shown as trace B of Fig. 5. Saturated fatty acid esters of polyethylene oxides such as those of lauric acid are excellent cohesive agents and show a medium level of fibre to fibre friction when applied to PP fibres [23]. PEO is also a common component in textile detergents and wetting agents [23].

3.5. Degumming of kenaf fibre due to chemical and enzymatic treatments

The kenaf fibres (technical fibres) are bundles of elementary fibres bonded together by “gums”, mainly hemicellulose, lignin and pectin. Chemical and enzyme treatments could remove some of the hemicellulose and other substances from the kenaf fibre and thus weaken the bonding between these elementary fibres, causing them to break up into smaller bundles and elementary fibres. This effect is known as degumming.

Table 5 shows the kenaf fibre weight loss due to these treatments. This weight loss was not all caused by degumming. Non-fibrous materials (“dusts”) attached to the fibre surface and small fibre fragments were also washed away in the treatments. The alkali treatment caused the most severe weight loss while the ethanol/water treatment the mildest.

Table 6 is a summary of the change of kenaf fibre diameter as a result of the chemical and enzymatic treatments applied. Not surprisingly, the alkali treatment, which is commercially used in the degumming of flax and other bast fibres by the textile industry, resulted in much more fibre separation (final fibre diameter 33 μm) than the other treatments (final fibre diameter between 51.2 and 54.8 μm). Despite the huge difference in the degumming effect, the alkali treatment did not provide a significantly different effect from the simple ethanol/water treatment in terms of the mechanical performance of the final composites, as shown in Fig. 1.

Table 5
Kenaf fibre weight loss due to chemical treatments.

Treatment	Weight loss (%)
Ethanol–water	3.20
Pectate lyase	5.58
Chelator	4.14
Laccase	4.96
Alkali	8.85

Table 6
Effect of treatment and carding on mean kenaf fibre diameter.

Treatment	Mean diameter (μm)	% Change in mean diameter
Untreated (before carding)	61.8 (48.2)	–
Untreated preform	60.2 (46.2)	–3
Ethanol/water	54.6 (44.0)	–12
Pectate lyase	51.2 (42.2)	–17
Laccase	54.8 (42.9)	–11
Alkali	33.0 (26.5)	–47

Standard deviation in bracket.

4. Conclusion

The relative benefits of two kenaf/polypropylene thermoplastic composite fabrication processes have been studied. The pre-treatment route, in which chemical or enzyme treatment is applied to the kenaf fibre before blending with the polypropylene fibre, is logistically simpler for the preform manufacturers and gives composite manufacturers greater flexibility. However, the post-treatment route, in which a blended preform is made from the kenaf and polypropylene fibres first followed by the required chemical or enzyme treatment, produced final composites with higher mechanical properties. This improvement has been attributed to the removal of the surface finish on the PP fibre by the chemical and enzyme treatments in the post-treatment route. The components of the surface finish can cause flaws to develop in the matrix of the final composites. The results call for spin finish formulations engineered for specific use in textile composites. The reduction of kenaf fibre diameter by the chemical and enzyme treatment due to “degumming” was not found to improve the flexural properties of the final composites produced.

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